

UTAMACS Portal — Product Improvement Recommendations

Context: Senior product owner research covering all 22 existing portal modules + Indian apartment market benchmarking. Market peers studied: NoBrokerHood, MyGate, ApnaComplex, Adda. Regulatory context: Telangana MACS Act 1995, TSRERA, DPDPA 2023, GST AAR on RWA maintenance, TDS §194C/194J, GHMC regulations.

Priority tiers: **P1** = critical gap / compliance risk / broken UX | **P2** = significant improvement | **P3** = polish / future

Already Implemented (this session)

Item	Module
Replace <code>alert()</code> with toast notifications	Polls
Vote result bars with % after casting	Polls
Exec "Close Poll" button (manual close)	Polls
<code>GET /api/v1/polls/:id/results</code> endpoint	Polls
<code>POST /api/v1/polls/:id/close</code> endpoint	Polls
Scheduled notices auto-publish via cron	Notices
In-app notifications on event create	Events
In-app notifications on poll create	Polls
In-app notifications on community post create	Community
<code>fanoutNotification()</code> utility + pref opt-out	Notifications
Religious + Maintenance category icons	Events
<code>comment_count</code> on community posts feed	Community
<code>community</code> type in notifications icon/link map	Notifications

1. Notices & Announcements

P1

- Acknowledgement tracking dashboard:** `requires_acknowledgement = true` notices have no exec view of who has/hasn't acknowledged. Add a read-receipt list per notice showing unit-wise status.
- Expiry auto-archive:** Notices past `expires_at` are still returned by the GET — filter them out or mark `status = 'archived'` via cron. Currently expired notices appear live.

- **Notice attachment signed URL:** Attachment stored in Supabase but the portal has no "View Attachment" button on the notices feed. Wire signed URL generation.
- **Announcement push (WhatsApp / SMS):** Urgent notices need an out-of-band channel. Integrate Gupshup or Twilio WhatsApp Business API. Exec can tick "Send WhatsApp" at publish time.
- **Notice delivery receipt:** Track per-user `is_read` at open time (not just acknowledgement). Useful for audit trail under DPDPA Article 9.

P2

- **Scheduled notices UI:** The `new.astro` form lets execs pick a scheduled time, but there is no "Scheduled" tab on the index page — execs can't see what's pending publication.
- **Notice categories as coloured pills on feed:** All categories currently render the same. Apply colour coding: Urgent = red, Financial = amber, General = blue, Maintenance = orange, Events = purple, Governance = teal.
- **Target audience filter (unit-type based):** `target_audience` column exists but the API doesn't filter by it. A notice targeted at "Flat owners only" reaches renters too.
- **Bulk resend:** Allow exec to resend a notice notification to members who haven't acknowledged.
- **Notice search:** Full-text search on title + body for members to find old announcements.

P3

- **Notice templates:** Common notice types (AGM reminder, water cut, maintenance shutdown) should have pre-filled templates.
- **Multilingual notices:** Telugu translation field alongside English body. Render toggle on feed.

2. Polls & Voting

P1

- **result_visibility respected in UI:** The DB column exists (`always` / `after_close` / `after_vote`) but the UI always shows results after voting. Enforce `after_close` — show a "Results will be visible after the poll closes" placeholder until then.
- **Multiple-choice poll type:** `poll_type = 'multiple_choice'` is a valid enum value but the vote API only stores one `option_id`. Need multi-select UI + array vote storage.
- **Rating poll type:** `poll_type = 'rating'` needs a 1–5 star or 1–10 slider UI.
- **Yes/No poll type:** Special two-button layout, larger tap targets.
- **One-vote-per-unit enforcement:** `one_vote_per_unit` column exists but the vote API checks only by `user_id`. If two residents from the same unit vote, both go through.
- **Poll notification to voters when results published:** After close, send a notification linking to results.

P2

- **Poll detail page `/portal/polls/[id]`:** Direct link from notification should land on the specific poll, not the index.
- **Poll comment thread:** Allow members to leave a comment alongside their vote (optional, configurable at creation).

- **CSV export of votes:** In addition to PDF, allow exec to download a CSV with (if not anonymous) voter name + unit + option chosen.
- **Poll archiving:** Old closed polls clutter the feed. Add an "Archived" tab or auto-hide polls closed > 30 days.
- **Scheduled poll start:** Similar to notices, allow a poll to open at a future `starts_at` rather than immediately on create.
- **Link poll to AGM:** Associate a poll with an AGM record so minutes auto-include the result.

P3

- **Anonymous voting audit trail:** Even when `is_anonymous = true`, allow the society's auditor role to verify unique-voter count without seeing who voted for what.
- **Poll duplication:** Copy an existing poll as a new draft.

3. Finance & Dues

P1

- **Member ledger / running balance:** Each resident needs a statement showing all dues, payments, and adjustments in chronological order with a running balance. Currently there's no way for a member to see their balance at a glance.
- **Payment gateway integration (Razorpay / Cashfree):** Online payment of maintenance dues is the single highest-value missing feature. Integrate Razorpay standard checkout. Store `razorpay_order_id` + `razorpay_payment_id` on payments. Webhook to confirm payment. RBI mandate: use only Indian payment aggregators.
- **Fund accounting — Sinking / Maintenance / Repair:** Telangana MACS Act requires separate accounting for Sinking Fund. Currently all expenses go into one bucket. Add `fund_type` column on expenses (`maintenance_fund`, `sinking_fund`, `repair_corpus`, `utility_pool`).
- **GST compliance for maintenance > ₹7,500/month:** RWA maintenance above ₹7,500/month/unit attracts 18% GST. The dues module must calculate and record GST on applicable units. Issuance of a GST invoice (already has pdfmake) is mandatory.
- **TDS on vendor payments (§194C / §194J):** Already partially built (admin TDS page). Ensure every expense > ₹30,000 single / ₹1L annual triggers a TDS deduction row. Generate Form 16A from portal.
- **Overdue interest calculation:** Auto-calculate interest on overdue dues (typically 12–18% p.a. as per by-laws). Cron adds interest rows monthly.
- **Expense approval workflow:** Currently expenses insert directly. Add a two-step flow: treasurer drafts → president/secretary approves. No payment without approval.

P2

- **Advance maintenance receipts:** Members who pay 12 months upfront get a single receipt valid for the year. Generate combined receipt PDF.
- **Partial payment handling:** A member who pays ₹3,000 against ₹5,000 due should have a `partially_paid` record, not be marked overdue.
- **Bank reconciliation view:** Upload a CSV bank statement; reconcile against payment records to identify unmatched credits.

- **Income tax receipt for flat owners:** Under IT Act §80C, RWA contributions are not deductible, but many owners ask for a receipt for Form 26AS reconciliation.
- **Budget vs. actual tracking:** Allow exec to create an annual budget per category and track actuals against it.
- **WhatsApp payment reminder:** Before the dues cron sends an in-app notif, also send a WhatsApp message with a direct payment link.

P3

- **CA/Auditor read-only login:** A special role for chartered accountant access to finance data without any write ability.
- **Annual financial report PDF:** Auto-generate the Receipts & Payments statement in the format required for MACS society annual returns.

4. Community Board

P1

- **Post images:** `community-images` Supabase bucket exists, and `/api/v1/community/posts/[id]/images.ts` exists, but the post creation form has no image upload UI. Wire it.
- **Moderation admin page:** Reports pile up in `community_reports` with no exec UI to review and act (remove post, warn user, dismiss report). Build `/portal/admin/moderation.astro`.
- **Edit own post:** Members can create but not edit. Add PATCH `/api/v1/community/posts/[id]` and an edit button visible to the post author.
- **Delete own post:** Author should be able to delete their own post (soft-delete `is_published = false`).
- **Pinned posts visible first:** API orders by `is_pinned DESC` but exec has no way to pin a post from the UI. Add a pin/unpin button (exec only).

P2

- **Rich text editor:** Replace the plain textarea in `new.astro` with a simple markdown or WYSIWYG editor (Trix, Quill lite, or just a toolbar over a textarea).
- **Comment notifications:** When someone comments on your post, you receive a notification.
- **Mention (@unit number):** Notify a specific resident via `@B-204` syntax in a post or comment.
- **Category icons on feed:** Visual icons per category (Alert = , Help = , Recommendation = , Lost_Found = ).
- **Post bookmarking:** Members can save posts to a personal reading list.
- **Community poll integration:** A "quick poll" widget embeddable inside a community post.

P3

- **Trending posts:** Surface posts with most reactions in the last 48 hours at the top.
- **Auto-expire Alert posts:** Category `Alert` posts older than 7 days auto-unpublish (configurable).

5. Events & RSVP

P1

- **QR check-in:** At event entry, a volunteer scans member's QR code (from profile) to mark attendance. Attendance vs. RSVP delta tracked.
- **Waitlist auto-promotion:** When a registered member cancels, the first person on the waitlist should be automatically promoted and notified.
- **Paid events / ticket collection:** `is_paid` and `ticket_price` columns exist but there is no payment flow. Wire Razorpay for ticket purchase. Generate a PDF ticket.
- **Event reminder notifications:** 24 hours and 1 hour before `starts_at`, push a reminder to all registered members.
- **Post-event feedback:** After `ends_at` passes, auto-send a feedback form link to attendees.

P2

- **Recurring events:** Monthly committee meetings, weekly yoga, etc. Add a `recurrence_rule` (RRULE) so one create generates multiple events.
- **Event banner image upload:** `event-banners` bucket exists but `new.astro` has no image upload. Wire it.
- **Event capacity warnings:** When 80% full, notify the event creator and show "Only X spots left" on the card.
- **iCal / Google Calendar export:** "Add to Calendar" button that generates an `.ics` file.
- **Event categories expanded:** Add Festive, Committee Meeting, Emergency Meeting, Workshop.
- **Guest pass:** RSVP for N attendees (not just 1). Control the per-unit guest limit.

P3

- **Event photo gallery:** After an event, exec uploads photos which appear in a linked gallery.
- **Volunteer sign-up:** Members can volunteer for event roles (setup, registration desk, cleanup).

6. Complaints

P1

- **SLA enforcement:** Complaints have no SLA timer. After 48 hours unacknowledged (configurable), auto-escalate to the next committee member.
- **Complaint attachment:** `complaint-attachments` bucket is provisioned but the complaint form has no file upload. Wire it.
- **Status change notifications:** When exec changes status (open → in_progress → resolved), notify the complainant automatically.
- **Public complaint counter:** Show open complaint count by category on the executive dashboard for accountability.

P2

- **Resolution notes field:** When marking resolved, exec must enter a resolution description. Currently resolution is silent.
- **Complaint categories:** Add structured categories (Water, Electrical, Lift, Security, Neighbour, Common Area, Sanitation) with sub-types.

- **Duplicate detection:** If two complaints describe the same location/category in 24 hours, prompt "a similar complaint already exists — merge?"
- **Resident satisfaction rating:** After resolution, ask the complainant to rate the resolution (1–5). Track committee performance.
- **Complaint history per unit:** On the member directory, show open complaint count per unit. Useful for tracking repeat issues.

P3

- **WhatsApp complaint submission:** Member sends a WhatsApp message to the society number, a bot creates a complaint automatically.
 - **Complaint heat map:** Visual grid of building/block showing complaint density by area.
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7. Facilities Booking

P1

- **Booking rules enforcement:** The `facility_rules` or `booking_rules` table (if it exists) is not enforced — members can book multiple consecutive slots or book on disallowed days. Add rule engine validation.
- **Cancellation refund logic:** If a member cancels a paid facility booking, the deposit refund process is manual. Automate the credit note.
- **Double-booking prevention:** Parallel POST requests for the same slot can create overlapping bookings if there's no DB-level uniqueness constraint.
- **Upcoming bookings on dashboard:** Member dashboard should show "Your next booking: Badminton Court, Tomorrow 6–7 PM".

P2

- **Facility photos:** `facility-photos` bucket exists but no image is displayed on the facility listing.
- **Booking lead time rules:** e.g., "Clubhouse must be booked minimum 48 hours in advance", "Maximum 2 bookings per unit per month".
- **Maintenance blackouts:** Exec marks a facility unavailable for a date range (painting, repair) and bookings are blocked.
- **Guest booking:** Allow booking a facility for a non-resident guest (name + phone logged).
- **Automated facility check-out reminder:** 15 minutes before booking end, WhatsApp the booker "Please wrap up and restore the facility."

P3

- **Facility utilisation analytics:** Which facilities are most booked, peak hours, average occupancy. Helps the committee justify new equipment.
 - **Booking waitlist:** If a slot fills, interested members join a waitlist and are auto-notified on cancellation.
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8. Visitor Management

P1

- **Guard mobile app / PWA:** Security guards currently have a desktop portal. A lightweight mobile-first PWA is essential for gate usage. Current `/portal/visitors` is desktop-only.
- **Pre-approval notification to guard:** When a resident pre-approves a visitor, the guard should receive an in-app notification at entry time.
- **Visitor photo capture:** Allow guards to photograph the visitor/vehicle at entry (Supabase Storage).
- **Frequent visitor quick-add:** For domestic help, drivers, and regular delivery agents, a "known visitor" list per unit for one-tap check-in.

P2

- **Visitor OTP verification:** On entry, guard triggers an OTP to the resident's phone. Resident approves via OTP. Eliminates phone tag.
- **Delivery package log:** Separate flow for package delivery (courier name, AWB, photo). Resident notified with "Package collected at gate."
- **Vehicle number plate log:** For vehicle entries, record number plate. Integrate with parivahan.gov.in API for verification (optional).
- **Contractor entry log:** Track contractor/vendor visits linked to open work orders.
- **Blacklist:** Exec can flag a person/vehicle as denied entry. Guard sees alert on check-in attempt.

P3

- **ANPR integration:** Automatic Number Plate Recognition for frequent vehicles (residents).
- **Daily visitor report to committee:** Cron sends a daily summary of visitor count, peak entry times to the exec.

9. HOTO Tracker

P1

- **RERA compliance packet:** Generate a single PDF "RERA Evidence Package" per unit containing all HOTO items, builder responses, and photo evidence. Required for TSRERA disputes.
- **Builder response tracking:** Currently there's a status field but no formal mechanism for the builder to upload a response or counter-response. Add a builder-response document upload.
- **HOTO item photos:** `complaint-attachments` or a dedicated bucket for HOTO photos is referenced but the HOTO item form has no photo upload UI.
- **Legal notice drafter:** When an item is overdue by 30 days, generate a draft legal notice (Word/PDF) addressed to the builder using project/builder details from system config.

P2

- **Unit-wise HOTO dashboard:** Each flat owner can log in and see only their unit's HOTO items. Currently the full HOTO list is exec-only.
- **Builder SLA vs. actual resolution chart:** Visual bar chart of committed vs. actual resolution dates by category (Civil, Electrical, Plumbing, Façade).
- **Cost recovery tracking:** Some HOTO resolutions cost the society money first. Track cost incurred and amount recovered from builder.

- **HOTO bulk upload:** Allow uploading a CSV of HOTO items from the handover inspection report.
- **Integration with Snags:** Close a HOTO item automatically when the linked snag is resolved.

P3

- **Telegram bot alerts:** Builder-facing Telegram bot where builder team can update status directly.
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10. AGM & Governance

P1

- **Minutes PDF auto-generation:** AGM minutes can be generated from the portal form but the output needs to be compliant with the MACS Act format (resolution numbering, quorum statement, attendance roll).
- **Quorum calculator:** Auto-check if quorum is met (e.g., 2/3 of members present) based on RSVP/attendance data linked from Events.
- **Proxy voting:** Under MACS rules, members unable to attend can assign proxy. Add proxy registration and recording.
- **AGM notice period:** Statutory 14-day notice required. The portal should warn if the AGM event is created less than 14 days out.

P2

- **Resolution tracking:** Each resolution passed at an AGM should be trackable (open / implemented / superseded). Link resolutions to action items.
- **Special resolution workflow:** Resolutions requiring 75% majority (MACS Act) need a special flag and quorum validation different from ordinary resolutions.
- **Agenda builder:** Drag-and-drop agenda items before the meeting; auto-include pending resolutions from the poll module.
- **Committee election module:** Coordinate elections for executive committee positions — nomination, voting, result declaration.

P3

- **Society registration documents vault:** Store all statutory filings (annual returns, audited accounts, RoCS submissions) with renewal reminders.
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11. Snag / Defect Tracking

P1

- **Snag photo upload:** `snag-attachments` Supabase bucket added in migration 048 but the snag form has no photo upload UI. Wire it.
- **Assignment to committee member:** Snags currently have no assignee. Add `assigned_to` (profile ID) so a committee member owns resolution.
- **Public snag dashboard:** A read-only view visible to all members showing open snag count by area (common areas, amenities, infrastructure). Builds trust.

P2

- **Snag SLA by category:** Different SLA days per snag category (Electrical = 24 hours, Civil = 7 days, Painting = 30 days).
- **Snag → vendor work order:** One-click creation of a vendor work order from an open snag. Auto-links the two records.
- **Snag bulk import from Excel:** Common during possession — hundreds of defects from a single inspection.
- **Resolution photo:** Require the assignee to upload a "before/after" photo before marking resolved.

P3

- **Snag heat map:** Building floor plan overlay showing snag locations by flat/area.
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12. Documents Library

P1

- **Real file upload:** Documents API has a signed-URL retrieval path but the upload flow needs to be end-to-end tested with actual multipart/form-data to Supabase Storage.
- **Version history:** When a document is updated (e.g., revised by-laws), keep old versions accessible with a version selector.
- **Access control per document:** Some documents (Financials, MoM) should be exec-only. Add a `visibility` enum: `all` / `exec` / `admin`.

P2

- **Full-text search:** Index document titles and descriptions for search.
- **Document expiry alerts:** Regulatory documents (PAN, GSTIN, BBMP NOC) have renewal dates. Remind exec 30 days before expiry.
- **Document categories:** Add Civil/Structural, Legal, Insurance, Regulatory, Society Rules, AGM Records.
- **Bulk download as ZIP:** Allow exec to download all documents in a category as a ZIP.

P3

- **OCR text extraction:** For scanned PDFs, extract text so they become searchable.
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13. Members Directory

P1

- **Unit occupancy status:** Is the flat owner-occupied, rented out, or vacant? Affects maintenance computation and voter eligibility.
- **Emergency contact per unit:** In a medical emergency, the guard needs a secondary contact. Currently not stored.
- **Member self-edit profile:** Members can only view their profile. Add ability to edit phone, email (with OTP verification), and profile photo.

- **DPDPA erasure request:** A member who exits the society must have a formal "data erasure" flow that anonymises PII while preserving financial records.

P2

- **Owner vs. tenant differentiation:** The `user_roles` table has no concept of "I own this flat but my tenant lives here." Add a `tenant_profiles` table linked to units.
- **Flat transfer workflow:** When a flat changes hands, formal transfer with document upload (sale deed), old owner data archival, new owner onboarding.
- **WhatsApp opt-in field:** Store WhatsApp number (may differ from registered phone). Required for WhatsApp Business API.
- **Member directory privacy:** Allow members to hide their phone/email from other members (visible only to exec/guard).

P3

- **Member anniversary/birthday:** Optional. Shows on committee dashboard; facilitates community bonding.

14. Parking Management

P1

- **Parking zone mapping:** Multi-level or segmented parking (Basement, Open, Stilt) with zone codes. Currently one pool.
- **Vehicle-to-slot assignment audit:** When a slot is reassigned, record the old owner + effective date. Disputes arise about historical assignments.
- **Visitor parking quota:** Track how many visitor slots are occupied in real time. Guard sees available count at gate.
- **RC/Insurance document upload:** `parking-docs` bucket is provisioned. Wire it in the parking registration form.

P2

- **EV charging slot management:** Mark specific slots as EV-charging-capable. Allocate on first-come basis with a waiting list.
- **Parking violation log:** Guard records unauthorised parking with photo; notif sent to unit; fine can be levied.
- **Guest parking pre-authorisation:** Resident reserves a visitor slot for a specific date/time window from the app.
- **Parking sticker generation:** Generate a printable PDF parking permit sticker with vehicle number and slot code.

P3

- **Sensor integration:** If future hardware sensors detect a vehicle in a slot, update occupancy in real time.

15. Vendors & Work Orders

P1

- **Work order status notifications:** When exec changes work order status, vendor and requester should receive notifications.
- **Vendor invoice upload:** After completing work, vendor uploads invoice; exec reviews and links to an expense record.
- **AMC (Annual Maintenance Contract) tracker:** Track vendor AMC contracts with `start_date`, `end_date`, `scope`, `amount`. Renewal reminders 30 days before expiry.
- **Vendor rating after work order close:** Ask the requester to rate the vendor (1–5). Build a vendor scorecard over time.

P2

- **Vendor payment history:** View all payments made to a vendor across work orders with running total.
- **Preferred vendor list:** Mark vendors as "approved" — only approved vendors can receive new work orders from non-exec members.
- **Work order template:** Common recurring work (lift maintenance, generator service) has pre-filled SOW templates.
- **Multi-vendor tender:** For large works, create a tender, invite multiple vendors to quote, compare quotes, award to one.

P3

- **Vendor onboarding portal:** A separate URL where new vendors self-register, upload GST registration, PAN, and insurance. Exec approves.
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16. Letters

P1

- **Letter delivery receipt:** Currently letters are generated as PDFs. Add a tracking mechanism: "Downloaded by member on [date]" — useful for legal purposes.
- **Letter templates library:** Common letters (NOC for property sale, rental NOC, dues clearance certificate) should be auto-fillable from member data.
- **Digital signature:** Society secretary signs letters with a digital stamp/signature image embedded in PDF.

P2

- **Letter archive per unit:** All letters issued to a unit accessible from the member's profile and from the unit record in the directory.
- **Bulk letter generation:** e.g., "Generate overdue dues notices for all 12 units with outstanding balances."
- **Letter approval workflow:** Draft → secretary review → president approval → issue. Currently letters can be issued without review.

P3

- **Registered post integration:** Partner with India Post API to dispatch physical letters where required (legal notices).
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17. Notifications Centre

P1

- **Unread badge on nav:** The notifications icon in the sidebar should show a red unread-count badge.
- **Real-time delivery:** Currently notifications are only fetched on page load / tab switch. Use Supabase Realtime to push new notifications without refresh.
- **Notification grouping:** Batch similar notifications ("3 new community posts", not 3 separate rows).
- **Digest mode:** For users who find notifications noisy, offer a daily digest email summarising the day's activities.

P2

- **Push notifications (web):** Register service worker + Web Push API so notifications reach members even when the portal is closed.
- **Email channel:** Send notification email via Resend for high-priority types (Urgent notice, overdue payment) even if the member hasn't opened the portal recently.
- **WhatsApp channel:** For members who opt in, deliver notifications via WhatsApp Business API.
- **Notification history filter:** Filter by type (Payments, Notices, Community...) — currently shows all in one list.
- **Mark all as read:** Currently only per-notification mark-read exists.

P3

- **Smart notification suppression:** If a member reads a notice directly, suppress the corresponding notification automatically.
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18. Domestic Help (Maids) Registry

P1 (module built; enhance)

- **Police verification document upload:** `maid-documents` bucket is provisioned. Wire it in the registration form.
- **Entry/exit log:** Guards log each domestic helper's check-in/check-out. Searchable history for security incidents.
- **Background check integration:** API integration with a background verification provider (AuthBridge, Aadhaar bridge) for automated screening.
- **Maid profile sharing:** If the same maid works in multiple units, avoid duplicate records. Link one maid profile to multiple units.

P2

- **Work schedule:** Track which units a maid services and on which days. Residents can see their maid's schedule and confirm.
- **Maid replacement/substitute tracking:** When a maid is on leave, track the substitute name temporarily.
- **Pay slip generation:** For societies that pay maids collectively, generate monthly pay slips.
- **Maid review / reliability rating:** Anonymous per-unit rating of the maid's reliability (opt-in).

P3

- **Group welfare schemes:** Track if the maid is enrolled in ESIC, PF (applicable in some organised setups).

19. Photo Gallery

P1 (module built; enhance)

- **Lazy loading:** All gallery images should use `loading="lazy"` and thumbnail variants. Currently full-resolution images are loaded immediately.
- **Image compression on upload:** Resize and compress images server-side before storing in Supabase to keep storage costs low.
- **Album grouping:** Group photos into albums (Diwali 2024, Pool Inauguration, Yoga Day). Currently flat list.
- **Member upload:** Allow any member to contribute photos to a shared album (with exec approval before publish).

P2

- **Download original:** Allow members to download original resolution photos.
- **Slideshow mode:** Full-screen slideshow with auto-advance.
- **Face privacy:** Before publishing event photos, allow members to request their face be blurred (DPDPA compliance).
- **Photo reactions:** Like/love reactions on photos.

P3

- **Video support:** Short videos (< 50 MB) alongside photos.
- **Year-in-review auto-album:** Cron auto-creates a year-end album from the top-reacted photos of the year.

20. Feedback

P1 (module built; enhance)

- **SLA for exec response:** Feedback submitted should receive an exec response within N days (configurable). Cron escalates overdue feedback.
- **Response visible to submitter:** Currently it's unclear if the exec's response is surfaced back to the member who submitted feedback. Wire the response display.

- **Anonymous feedback analysis:** Aggregate anonymous feedback by category for committee without revealing identities.

P2

- **Periodic feedback surveys:** Exec creates a multi-question survey (NPS + open text) and sends to all members. Results aggregated automatically.
- **Feedback categories:** Cleanliness, Security, Committee Performance, Amenities, Communication. Helps trend analysis.
- **Public satisfaction score:** Show society's average member satisfaction score on the exec dashboard.

P3

- **Benchmarking:** Compare satisfaction score month-over-month.
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21. Analytics / Reports Hub

P1

- **Financial summary report:** Monthly P&L — collections vs. expenses vs. outstanding. Downloadable PDF for committee review.
- **Occupancy report:** How many flats are owner-occupied vs. rented vs. vacant. Important for voter lists and maintenance load.
- **Collection efficiency:** Percentage of dues collected on time, overdue rate by month.

P2

- **Complaint resolution time chart:** Average days-to-resolve by category over rolling 6 months.
- **Event attendance analytics:** RSVP vs. actual attendance ratio per event type.
- **Vendor spend analysis:** Top 5 vendors by expenditure; spend by category (Civil, Electrical, Landscaping).
- **Facility utilisation:** Booking hours per facility per month.
- **Member engagement score:** Polls voted, events attended, community posts — a simple engagement index per unit.

P3

- **Predictive maintenance calendar:** Based on equipment age and past expense history, suggest when lifts, generators, pumps need servicing.
 - **Export to Excel/CSV:** All reports downloadable as spreadsheets in addition to PDF.
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22. Admin (RBAC, Audit, Staff, TDS)

P1

- **Audit log viewer:** `audit_logs` table exists but there's no portal page to search/filter it. Exec/admin should be able to search by user, resource type, date range.

- **Session management:** Show all active sessions for a user and allow remote logout (DPDPA requirement for data access control).
- **Staff account lifecycle:** Security guard and vendor accounts need formal deactivation when the person leaves. Currently it's a manual Supabase operation.
- **RBAC change audit:** All role grants/revocations must be logged in audit_logs with the grantor's identity.

P2

- **Feature flag UI:** Currently feature flags are database rows. Provide a toggle UI so admin can enable/disable modules per society without SQL.
- **System health dashboard:** Show Supabase connection status, recent cron run results, upload queue depth, storage usage.
- **TDS certificate generation (Form 16A):** Download Form 16A for each vendor for the financial year.
- **Bulk user import:** Upload a CSV of new members for batch onboarding.

P3

- **Multi-society support:** The schema already has `society_id` throughout. A future admin role can manage multiple societies from one login.

New Modules — High Priority

NM-1: Water / Tanker Management (P1)

Indian apartment societies routinely order water tankers during HMWSSB supply interruptions. Residents currently call the committee directly.

Features: Tanker booking with vendor, volume, and cost tracking; notify all members when a tanker is ordered; per-unit water consumption log if sub-meters exist; bore-well pump maintenance log; water quality test record with upload.

NM-2: Security Patrol Log (P1)

Guards need to log their patrol checkpoints and any incidents observed during rounds. Currently entirely manual.

Features: Guard taps a QR sticker at each patrol checkpoint (lift, gate, parking, terrace); timestamp + location logged; incident flag with photo; daily patrol report auto-emailed to exec; patrol gaps alerted (if no checkpoint scanned in 2 hours during night duty).

NM-3: AMC / Planned Maintenance Calendar (P1)

Lifts, generators, pumps, and CCTV are maintained on annual contracts. Due dates are tracked in spreadsheets, causing missed services.

Features: AMC register for each equipment item (vendor, contract start/end, value, scope); preventive maintenance schedule (monthly, quarterly, half-yearly); cron sends reminder 30 days before each service due; service completion log with engineer name and remarks; link to vendor record and expense entry.

NM-4: Move-In / Move-Out (P2)

Flat handovers involve NOC clearance, utility meter reading, and damage deposit. No structured workflow exists.

Features: Resident initiates move-out request; exec checklist (dues cleared, parking surrendered, maid access removed); meter reading recorded at move-out; damage deposit refund approval workflow; flat re-listed as vacant until new owner/tenant registered; move-in checklist for new resident.

NM-5: GST / TDS Compliance Dashboard (P1)

UTAMACS must file GSTR-1, GSTR-3B and TDS returns quarterly. Currently done manually from spreadsheet data.

Features: Consolidate all maintenance income (GST applicable above ₹7,500/unit/month); generate GSTR-1 data export; consolidate TDS deducted on vendor payments; generate Form 26Q data export; track GST registration status and return due dates; alert exec 7 days before filing deadline.

NM-6: Delivery / Package Tracking (P2)

E-commerce deliveries to gated communities are a daily friction point. Guards receive packages on behalf of residents.

Features: Guard logs package (courier, AWB, photo); resident notified immediately; resident acknowledges pickup; report of packages held for > 24 hours; integration with major courier tracking APIs (Delhivery, BlueDart, Ekart) for auto-log on SMS.

NM-7: Tenant KYC / Police Verification (P1)

Under Hyderabad City Police guidelines and TSRERA, rental tenants must be police-verified. Society must maintain a tenant register.

Features: Owner registers tenant with ID proof (Aadhaar masked), photo, emergency contact; auto-generate police verification letter for submission to local station; Aadhaar verification (DigiLocker API); tenant lease document upload; notify exec on new tenant registration for approval.

NM-8: Utility Monitoring (P2)

Common area electricity and water consume a large portion of the society's expense. Proactive monitoring reduces cost.

Features: Monthly common area electricity unit entry (or smart meter integration); benchmark per-month usage; alert if month-on-month exceeds threshold; water consumption from tanker + borewell log; solar

rooftop generation log (if applicable); utility expense auto-linked to finance module.

NM-9: School Bus Coordination (P3)

Many apartment societies in Hyderabad run school bus pooling. Currently managed by parent WhatsApp groups.

Features: Bus route and stop registration; student registration (unit, school, class, timing); parent receives route deviation alert (GPS integration optional); monthly fee collection via Finance module; driver profile with license and background check; emergency SOS button on bus.

Indian Market Context: Key Differentiators to Build

These items are absent from or weak in NoBrokerHood / MyGate but highly valued in Telangana-specific societies:

1. **Telugu language support:** NavBar, notices, and key forms in Telugu. Significant portion of non-English-comfortable residents.
 2. **TSRERA-ready evidence packaging:** Automatic generation of a RERA complaint packet (photos + correspondence timeline + builder SLA + cost incurred).
 3. **GHMC property tax integration:** Help members verify their property tax status and due amounts from the society portal (screen-scrape or GHMC API if available).
 4. **HMWSSB water bill tracking:** Track municipality water supply hours per building block; correlate with tanker spend.
 5. **Telangana MACS Act compliance:** Statutory meeting frequency, notice periods, quorum requirements, register maintenance — all built into workflows rather than left to exec knowledge.
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